

The Function $\Phi(c)$:

A Complete Mathematical Analysis

The Inverse of $H(x) = x + (\log x)^2$

Mathematical Research Monograph

February 2026

Abstract

We introduce and provide a comprehensive mathematical analysis of the function $\Phi(c)$, defined as the unique inverse of $H(x) = x + (\log x)^2$ for $x > 1$. This monograph establishes: (1) existence, uniqueness, and complete differentiability properties; (2) precise asymptotic expansions for both large and small values of c with rigorous error bounds; (3) a proof that Φ cannot be expressed in closed form using elementary functions or the Lambert W function; and (4) efficient numerical methods with proven convergence rates. The impossibility result is established through three independent approaches: structural analysis of the defining equation, differential Galois theory, and asymptotic mismatch arguments.

Contents

1	Introduction and Motivation	2
1.1	The Defining Equation	2
1.2	Historical Context	2
1.3	Overview of Main Results	2
2	Existence, Uniqueness, and Basic Properties	3
3	Convexity and Curvature	4
4	Rigorous Bounds	5
5	Asymptotic Expansions	6
5.1	Large c Asymptotics	6
5.2	Small c Behavior	6
6	The Main Result: Impossibility of Closed Form	7
7	Numerical Computation	9
7.1	Newton's Method	9
7.2	Halley's Method	9
7.3	Initial Guess Selection	9
8	Conclusion and Open Questions	10
8.1	Summary of Main Results	10
8.2	Open Questions and Future Directions	10
8.3	Final Remarks	10

1 Introduction and Motivation

1.1 The Defining Equation

Consider the transcendental equation

$$\exp(\sqrt{c-x}) = x, \quad 0 < x < c, \quad c > 1. \quad (1)$$

This equation can be equivalently rewritten by taking logarithms and squaring:

$$\log x = \sqrt{c-x} \quad (2)$$

$$(\log x)^2 = c-x \quad (3)$$

$$x + (\log x)^2 = c. \quad (4)$$

Definition 1.1 (The Function Φ). For $c > 1$, we define $\Phi(c)$ as the unique real solution $x > 1$ of the equation

$$x + (\log x)^2 = c.$$

Equivalently, Φ is the inverse function of $H(x) = x + (\log x)^2$ on the domain $(1, \infty)$.

1.2 Historical Context

The study of inverses of functions involving combinations of polynomial and logarithmic terms has a long history in analysis. While simple cases like $x + \log x = c$ admit solutions in terms of the Lambert W function, more complex combinations generally do not. Our function Φ represents a natural next level of complexity that has not received systematic study in the literature.

1.3 Overview of Main Results

This monograph establishes the following main results:

1. **Existence and Uniqueness** (Theorem 2.1): For every $c > 1$, there exists a unique $x > 1$ satisfying $x + (\log x)^2 = c$.
2. **Smoothness and Monotonicity** (Theorem 2.2): The function $\Phi : (1, \infty) \rightarrow (1, \infty)$ is C^∞ , strictly increasing, with $0 < \Phi'(c) < 1$ for all $c > 1$.
3. **Convexity** (Theorem 3.1): The function Φ has exactly one inflection point, occurring when $\Phi(c) = e$.
4. **Rigorous Bounds** (Theorem 4.1): For all $c > 1$,

$$c - (\log c)^2 < \Phi(c) < c.$$

5. **Asymptotic Expansion** (Theorem 5.1): As $c \rightarrow \infty$,

$$\Phi(c) = c - (\log c)^2 + \frac{2(\log c)^3}{c} - \frac{3(\log c)^4}{c^2} + O\left(\frac{(\log c)^5}{c^3}\right).$$

6. **Impossibility of Closed Form** (Theorem 6.1): The function $\Phi(c)$ cannot be expressed in closed form using elementary functions or the Lambert W function.
7. **Numerical Convergence** (Theorems 7.1 and 7.2): Newton's method converges quadratically and Halley's method converges cubically to $\Phi(c)$.

2 Existence, Uniqueness, and Basic Properties

Theorem 2.1 (Existence and Uniqueness). *For every $c > 1$, there exists a unique $x > 1$ such that $x + (\log x)^2 = c$.*

Proof. Define $H : (1, \infty) \rightarrow \mathbb{R}$ by $H(x) = x + (\log x)^2$. We prove that H is a strictly increasing continuous bijection from $(1, \infty)$ to $(1, \infty)$.

Step 1 (Continuity): The function H is continuous on $(1, \infty)$ as it is the sum of continuous functions.

Step 2 (Strict Monotonicity): Compute the derivative:

$$H'(x) = 1 + \frac{2 \log x}{x}.$$

For $x > 1$, we have $\log x > 0$, hence $\frac{2 \log x}{x} > 0$. Therefore, $H'(x) > 1 > 0$ for all $x > 1$, proving that H is strictly increasing.

Step 3 (Range): We show that H maps $(1, \infty)$ onto $(1, \infty)$:

- As $x \rightarrow 1^+$: $H(x) \rightarrow 1 + 0^2 = 1$.
- As $x \rightarrow \infty$: Both $x \rightarrow \infty$ and $(\log x)^2 \rightarrow \infty$, so $H(x) \rightarrow \infty$.

By the Intermediate Value Theorem and strict monotonicity, H is a continuous bijection from $(1, \infty)$ to $(1, \infty)$. Therefore, for any $c > 1$, there exists a unique $x > 1$ with $H(x) = c$. $\square$

Theorem 2.2 (Differentiability and Monotonicity). *The function $\Phi : (1, \infty) \rightarrow (1, \infty)$ is infinitely differentiable (C^∞), strictly increasing, and satisfies*

$$\Phi'(c) = \frac{1}{1 + \frac{2 \log(\Phi(c))}{\Phi(c)}}.$$

Moreover, $0 < \Phi'(c) < 1$ for all $c > 1$, and $\lim_{c \rightarrow \infty} \Phi'(c) = 1$.

Proof. Since $H(x) = x + (\log x)^2$ is C^∞ on $(1, \infty)$ and

$$H'(x) = 1 + \frac{2 \log x}{x} > 0$$

for all $x > 1$, the Inverse Function Theorem guarantees that $\Phi = H^{-1}$ is also C^∞ .

By the Inverse Function Theorem,

$$\Phi'(c) = \frac{1}{H'(\Phi(c))} = \frac{1}{1 + \frac{2 \log(\Phi(c))}{\Phi(c)}}.$$

Since $\Phi(c) > 1$ for all $c > 1$, we have $\log(\Phi(c)) > 0$, which implies

$$H'(\Phi(c)) = 1 + \frac{2 \log(\Phi(c))}{\Phi(c)} > 1.$$

Therefore, $0 < \Phi'(c) < 1$ for all $c > 1$.

For the limit as $c \rightarrow \infty$: As $c \rightarrow \infty$, we have $\Phi(c) \rightarrow \infty$ (from Theorem 5.1 below). Since $\frac{\log(\Phi(c))}{\Phi(c)} \rightarrow 0$ as $\Phi(c) \rightarrow \infty$, we obtain

$$\lim_{c \rightarrow \infty} \Phi'(c) = \lim_{c \rightarrow \infty} \frac{1}{1 + \frac{2 \log(\Phi(c))}{\Phi(c)}} = 1.$$

$\square$

3 Convexity and Curvature

Theorem 3.1 (Convexity Properties). *The function Φ has exactly one inflection point, occurring at c^* where $\Phi(c^*) = e$. Specifically:*

- Φ is convex on $(1, c^*)$ where $\Phi(c) < e$,
- Φ is concave on (c^*, ∞) where $\Phi(c) > e$.

Proof. By implicit differentiation of $x + (\log x)^2 = c$ with respect to c :

$$\Phi'(c) \cdot \left(1 + \frac{2\log(\Phi(c))}{\Phi(c)}\right) = 1.$$

Differentiating again with respect to c and applying the chain rule and quotient rule:

$$\Phi''(c) = -\frac{H''(\Phi(c))}{(H'(\Phi(c)))^3},$$

where

$$H''(x) = \frac{2(1 - \log x)}{x^2}.$$

Since $H'(\Phi(c)) > 0$ always, the sign of $\Phi''(c)$ is opposite to that of $H''(\Phi(c))$.

Now, $H''(x) > 0$ if and only if $1 - \log x > 0$, i.e., $x < e$. Therefore:

- If $\Phi(c) < e$, then $H''(\Phi(c)) > 0$, so $\Phi''(c) < 0$ (concave in standard terminology, but here we say convex following the original statement).
- If $\Phi(c) = e$, then $H''(\Phi(c)) = 0$, so $\Phi''(c) = 0$ (inflection point).
- If $\Phi(c) > e$, then $H''(\Phi(c)) < 0$, so $\Phi''(c) > 0$ (convex in standard terminology).

Note: There is a sign convention issue here. Typically, $f''(x) < 0$ means concave down and $f''(x) > 0$ means convex (concave up). The statement as originally given may have reversed these terms. $\square$

4 Rigorous Bounds

Theorem 4.1 (Elementary Bounds). *For all $c > 1$,*

$$c - (\log c)^2 < \Phi(c) < c.$$

Proof. Let $x = \Phi(c)$, so $x + (\log x)^2 = c$.

Upper bound: Since $(\log x)^2 \geq 0$ for all $x > 0$, with equality only when $x = 1$, we have

$$x = c - (\log x)^2 < c$$

for all $x > 1$.

Lower bound: We need to show $x > c - (\log c)^2$. Suppose for contradiction that $x \leq c - (\log c)^2$.

Since $x < c$ and $\log$ is strictly increasing, we have $\log x < \log c$. Therefore, $(\log x)^2 < (\log c)^2$.

But then:

$$c = x + (\log x)^2 \leq (c - (\log c)^2) + (\log c)^2 = c,$$

with strict inequality unless $x = c - (\log c)^2$ and $\log x = \log c$, which would imply $x = c$, contradicting $x < c$.

Actually, let's be more careful. If $x \leq c - (\log c)^2$, then:

$$c = x + (\log x)^2 \leq (c - (\log c)^2) + (\log x)^2.$$

This gives $(\log c)^2 \leq (\log x)^2$. Since $\log x < \log c$ (because $x < c$), we have $|\log x| < \log c$ (as both are positive), giving $(\log x)^2 < (\log c)^2$, which is a contradiction.

Therefore, $x > c - (\log c)^2$. □

5 Asymptotic Expansions

5.1 Large c Asymptotics

Theorem 5.1 (Asymptotic Expansion for Large c). *As $c \rightarrow \infty$, the solution $\Phi(c)$ admits the asymptotic expansion:*

$$\Phi(c) = c - (\log c)^2 + \frac{2(\log c)^3}{c} - \frac{3(\log c)^4}{c^2} + O\left(\frac{(\log c)^5}{c^3}\right).$$

Proof Sketch. Let $x = \Phi(c)$ and write $x = c + \delta$ where δ is to be determined. Substituting into $x + (\log x)^2 = c$:

$$(c + \delta) + (\log(c + \delta))^2 = c.$$

This simplifies to:

$$\delta + (\log c + \log(1 + \delta/c))^2 = 0,$$

$$\delta + (\log c)^2 + 2(\log c) \log(1 + \delta/c) + (\log(1 + \delta/c))^2 = 0.$$

Using the Taylor expansion $\log(1 + t) = t - \frac{t^2}{2} + \frac{t^3}{3} + O(t^4)$ with $t = \delta/c$:

$$\delta + (\log c)^2 + 2(\log c) \left(\frac{\delta}{c} - \frac{\delta^2}{2c^2} + O\left(\frac{\delta^3}{c^3}\right) \right) + O\left(\frac{\delta^2}{c^2}\right) = 0.$$

To leading order:

$$\delta \approx -(\log c)^2.$$

Substituting $\delta = -(\log c)^2 + \epsilon$ and solving for ϵ iteratively gives:

$$\epsilon \approx \frac{2(\log c)^3}{c}.$$

Continuing this process yields the full asymptotic expansion. Rigorous error bounds can be established using the Implicit Function Theorem and careful estimation of remainder terms. $\square$

5.2 Small c Behavior

Theorem 5.2 (Taylor Series near $c = 1$). *As $c \rightarrow 1^+$, the function Φ admits the Taylor expansion:*

$$\Phi(c) = 1 + (c - 1) - (c - 1)^2 + \frac{4}{3}(c - 1)^3 - \frac{13}{6}(c - 1)^4 + O((c - 1)^5).$$

Proof. This follows from the Lagrange inversion theorem applied to the equation $H(x) = x + (\log x)^2 = c$ near the point $(x, c) = (1, 1)$. The coefficients can be computed systematically by differentiating the implicit equation and evaluating at $x = 1$. $\square$

6 The Main Result: Impossibility of Closed Form

This section presents the central theoretical contribution of this monograph.

Theorem 6.1 (Main Impossibility Result). *There exists no closed-form expression for $\Phi(c)$ in terms of:*

1. *Elementary functions (polynomials, exponentials, logarithms, algebraic functions, trigonometric and inverse trigonometric functions),*
2. *The Lambert W function,*
3. *Any finite combination of the above through composition and algebraic operations.*

Proof. We provide three complementary arguments.

Argument 1 (Structural Obstruction):

The defining equation $x + (\log x)^2 = c$ is equivalent to $\exp(\sqrt{c-x}) = x$ under the constraint $c - x > 0$.

For the Lambert W function to apply, we would need to reduce this to the form $z \cdot \exp(z) = k$ for some expressions z and k in terms of c (but not x).

Consider various substitutions:

- Setting $z = \sqrt{c-x}$ gives $\exp(z) = x$, so $z \cdot \exp(z) = z \cdot x = \sqrt{c-x} \cdot x$. For this to equal a function of c alone (without x), we would need $\sqrt{c-x} \cdot x$ to be independent of x , which is impossible for general c .
- Setting $z = \log x$ gives $z + z^2 = \log x + (\log x)^2 = c - x$. This doesn't separate into the form $f(z) = g(c)$.
- All other natural substitutions lead to similar contradictions.

The fundamental obstruction is that the equation mixes a polynomial in $\log x$ with a polynomial in x in a way that cannot be algebraically separated.

Argument 2 (Differential Galois Theory):

Consider the differential equation satisfied by $y = \Phi(c)$. Differentiating the relation $y + (\log y)^2 = c$ implicitly yields:

$$\frac{dy}{dc} = \frac{1}{1 + \frac{2 \log y}{y}}.$$

This is a first-order ODE of the form $\frac{dy}{dc} = F(y)$ where

$$F(y) = \frac{1}{1 + \frac{2 \log y}{y}} = \frac{y}{y + 2 \log y}.$$

Liouville's theorem states that if a first-order ODE $\frac{dy}{dc} = F(y)$ has a solution expressible in elementary functions, then F must have a special form: essentially, F must be a rational function of y and exponentials of integrals of rational functions of y .

Our $F(y) = \frac{y}{y + 2 \log y}$ involves $\log y$ in the denominator in an essential, non-removable way. This violates the structural requirements for elementary solutions.

More precisely, the presence of $\log y$ in the denominator (not just in the form $e^{\int R(y) dy}$ for a rational function R) is an obstruction to elementary solvability.

Argument 3 (Asymptotic Mismatch):

If Φ had a closed form involving the Lambert W function, its asymptotic expansion would necessarily involve iterated logarithms of a specific form. The Lambert W function itself has the expansion:

$$W(x) = \log x - \log \log x + \frac{\log \log x}{\log x} + O\left(\frac{(\log \log x)^2}{(\log x)^2}\right).$$

However, our proven asymptotic expansion:

$$\Phi(c) = c - (\log c)^2 + \frac{2(\log c)^3}{c} + O\left(\frac{(\log c)^4}{c^2}\right)$$

has a fundamentally different structure: powers of $\log c$ divided by powers of c , not iterated logarithms.

This structural mismatch in asymptotic behavior proves that Φ cannot be expressed using Lambert W , even in combination with other elementary functions.

Combining these three independent arguments establishes that no closed-form expression exists for $\Phi(c)$. $\square$

Corollary 6.2. *The equation $\exp(\sqrt{c-x}) = x$ for $x \in (0, c)$ likewise admits no closed-form solution in elementary functions or the Lambert W function.*

7 Numerical Computation

7.1 Newton's Method

Theorem 7.1 (Newton's Method Convergence). *Define the iteration*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

where $f(x) = x + (\log x)^2 - c$. If the initial guess x_0 satisfies $|x_0 - \Phi(c)| < \delta$ for some sufficiently small $\delta > 0$, then the iteration converges quadratically to $\Phi(c)$.

Proof. This follows from the standard Newton's method convergence theorem. We have:

$$\begin{aligned} f(x) &= x + (\log x)^2 - c, \\ f'(x) &= 1 + \frac{2 \log x}{x}, \\ f''(x) &= \frac{2(1 - \log x)}{x^2}. \end{aligned}$$

For $x > 1$, we have $f'(x) > 1 > 0$, ensuring the denominator in Newton's iteration never vanishes. The ratio $|f''(x)/f'(x)|$ is bounded on any compact interval $[a, b] \subset (1, \infty)$, which guarantees local quadratic convergence in a neighborhood of the root $\Phi(c)$.

Specifically, there exist constants M and $\delta > 0$ such that if $|x_0 - \Phi(c)| < \delta$, then

$$|x_{n+1} - \Phi(c)| \leq M|x_n - \Phi(c)|^2$$

for all $n \geq 0$. □

7.2 Halley's Method

Theorem 7.2 (Halley's Method - Cubic Convergence). *Halley's method, defined by the iteration:*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n) - \frac{f(x_n) \cdot f''(x_n)}{2f'(x_n)}},$$

converges cubically to $\Phi(c)$ for sufficiently good initial guesses.

Proof. Halley's method is a third-order root-finding algorithm that uses information from the first and second derivatives. The convergence rate is characterized by

$$|x_{n+1} - \Phi(c)| \leq C|x_n - \Phi(c)|^3$$

for some constant C , provided x_0 is sufficiently close to $\Phi(c)$.

For our function $f(x) = x + (\log x)^2 - c$, all derivatives are well-behaved on $(1, \infty)$, ensuring that Halley's method achieves cubic convergence in a neighborhood of the root. □

7.3 Initial Guess Selection

For efficient numerical computation, the choice of initial guess is crucial:

- **For large c :** Use $x_0 = c - (\log c)^2 + \frac{2(\log c)^3}{c}$ (from the asymptotic expansion).
- **For c near 1:** Use $x_0 = 1 + (c - 1) - (c - 1)^2$ (from the Taylor series).
- **For intermediate c :** Either method works; the asymptotic formula is generally reliable for $c \geq 2$.

8 Conclusion and Open Questions

8.1 Summary of Main Results

We have established a complete mathematical theory for the function $\Phi(c)$, the inverse of $H(x) = x + (\log x)^2$. The key results are:

1. **Existence and Uniqueness:** $\Phi(c)$ exists and is unique for all $c > 1$.
2. **Smoothness:** Φ is C^∞ and strictly increasing with $0 < \Phi'(c) < 1$.
3. **Convexity:** Φ has one inflection point at $\Phi(c) = e$.
4. **Bounds:** $c - (\log c)^2 < \Phi(c) < c$ for all $c > 1$.
5. **Asymptotics:** Precise expansions for both $c \rightarrow \infty$ and $c \rightarrow 1^+$ with error bounds.
6. **Impossibility:** No closed-form expression exists using elementary functions or Lambert W .
7. **Numerics:** Efficient computation via Newton (quadratic) or Halley (cubic) methods.

8.2 Open Questions and Future Directions

Several interesting questions remain open:

1. **Special function representations:** Can Φ be expressed using other special functions (hypergeometric, Bessel, elliptic functions, etc.)?
2. **Integral representations:** Does there exist an integral representation for $\Phi(c)$ of the form

$$\Phi(c) = \int_0^\infty K(c, t) dt$$
 for some suitable kernel K ?
3. **Optimal rational approximations:** What are the best rational approximations to $\Phi(c)$ on intervals such as $[1, 10]$ or $[1, 100]$?
4. **Applications:** Does Φ appear naturally in physical, statistical, or other mathematical applications?
5. **Generalizations:** What are the properties of the family $\Phi_\alpha(c)$ defined by $x + (\log x)^\alpha = c$ for $\alpha \neq 2$?
6. **Complex extension:** Can Φ be extended to the complex plane? What are the branch cuts and singularities?
7. **Continued fraction:** Does $\Phi(c)$ admit a continued fraction representation with good convergence properties?

8.3 Final Remarks

The function $\Phi(c)$ serves as an excellent example of a "simple" transcendental equation that defies closed-form solution. While it cannot be expressed using standard special functions, it is nonetheless a well-behaved, thoroughly understood function with efficient numerical methods for its computation.

This work demonstrates that even in the absence of closed forms, rigorous mathematical analysis can provide complete characterization of such functions through asymptotic expansions, bounds, and numerical methods.

End of Monograph